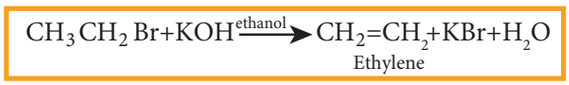
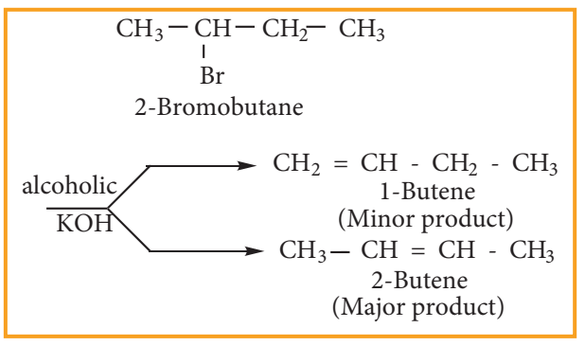
releasing a halogen attached to a α carbon and a hydrogen to a β carbon of halo alkane. This reaction is called β elimination reaction. (dehydrohalogenation).



Some haloalkanes yield a mixture of olefins in different amounts. It is explained by Saytzeff's Rule, which states that 'In a dehydrohalogenation reaction, the preferred product is that alkene which has more number of alkyl groups attached to the doubly bonded carbon (more substituted double bond is formed)

Example



Elimination reactions may proceed through two different mechanisms namely E_1 and E_2

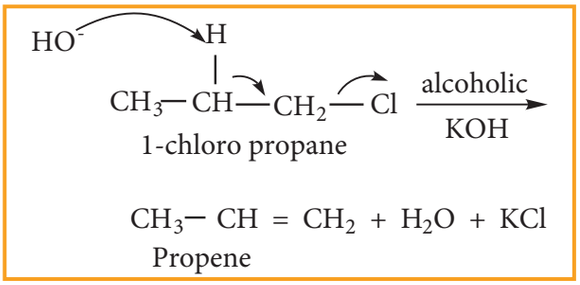
E_2 reaction mechanism

Elimination
E_2 → bimolecular
(Second order)

The rate of E_2 reaction depends on the concentration of alkyl halide and base

Rate = k [alkyl halide][base]

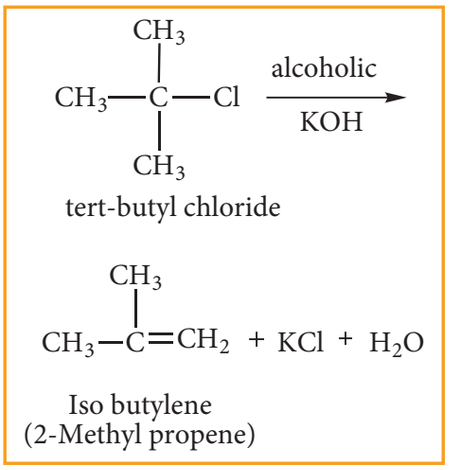
It is therefore, a second order reaction. Generally primary alkyl halide undergoes this reaction in the presence of alcoholic KOH. It is a one step process in which the abstraction of the proton from the β carbon and expulsion of halide from the ∝ carbon occur simultaneously. The mechanism is shown below.



E_1 reaction mechanism

Elimination
E_1 → Unimolecular

Generally, tertiary alkyl halide which undergoes elimination reaction by this mechanism in the presence of alcoholic KOH. It follows first order kinetics. Let us consider the following elimination reaction.



Step - 1 Heterolytic fission to yield a carbocation

Unit 14 - jagan.indd 237 22-01-2021 13:04:59